

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 18

(1) Research Program/Project/Study Title: Variation in environmental characteristics of waters among Oyster and Mussel culture areas in Pilar Bay, Capiz	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center						
(4) Site of implementation								
Municipality/City:		Pilar Bay						
Province:		Capiz						
Region:		VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research
Research:	Development:							
☐ Basic	☐ Pilot Testing							
☐ Applied	☐ Tech. Promotion/Commercialization							

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH		Revision No.	00
				Effective Date	
				Page	Page 2 of 18

		(external agency)
(7) Sector/Commodity Oyster and Mussel		(8) Discipline/Field of Specialization
(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.) (9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.) Oysters and Mussel are one of the important commodities being cultured and produced in the Province of Capiz. Thus, the present study aims to assess and compare the water characteristics based on months and between each site as well as determine the zonation of oyster and mussel culture areas within Pilar Bay, Capiz. The study areas are located in Barangay Binaobawan in Pilar, Brgy. San Pedro in Ponedvedra, Brgy. Buntod, Agojo, and Bantigue in Panay, and Brgy. Madulano, and Pinamihagan in President Roxas. The municipalities that encompass Pilar Bay which is considered as one of the important fishing grounds in the province. In situ, physicochemical parameters (temperature, salinity, dissolved oxygen, pH, ORP, and conductivity) were measured from 7 sampling sites monthly from January to October 2023 and statistically analyzed using SPSS. Map showing the geotagged oysters and mussels in the sites were generated. Furthermore, results revealed that all six physicochemical parameters tested significantly ($p < 0.05$) different in terms of months and sites except Dissolved Oxygen and Salinity. This variation in the environmental characteristics of waters in the oyster and mussel culture area in Pilar Bay could be attributed to freshwater influx due to the frequent precipitation flooding and tidal variation. Site suitability was also assessed, and the result showed that among the sites, Bantigue turned out to be the most suitable site for oyster culture while Buntod and Agojo were the least suitable however, Buntod and Agojo were the most suitable site for mussel farming. Other sites were moderately suitable for both oyster and mussel farming. Keywords: <i>Pilar Bay, Oyster, Mussel, Physico-chemical parameters, environmental characteristics</i>		
(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews) Shellfish aquaculture, with a particular emphasis on the cultivation of oysters and mussels, serves as an important mechanism for economic development in the province of Capiz, Philippines. These two shellfish species are regarded as fundamental and most cultivated aquaculture fishery commodities mainly cultivated for		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 18

their meat and consumption. However, it is now on a commercial scale, providing livelihood to fishermen and their families. Philippine Green Mussels (*Perna viridis*) locally known as “tahong” and slipper-shaped oysters (talabang tsinelas) can be marketed locally and globally and their production requires minimum capital investment.

Due to their high protein value, easy and inexpensive breeding, and fast growth, tahong and talaba products have intensified in recent years (Echapare et. al. 2019). Over ten years (2011-2020), the shellfish industry showed an annual production averaging 49,656.65 MT (PSA, 2020), or a 4.69% yearly average percentage increase for the last ten years. The data showed that one of the top-producing regions is Region VI (21,388.62 MT). Production areas or provinces from the top producing regions include Aklan, Capiz, and Iloilo as reported by Regional Shellfish Focal Persons of the Bureau of Fisheries and Aquatic Resources. However, production declined to 20.1 MT in 2008. The decline in oyster production was at 4.90% and 2.13 % for the provinces of Negros Occidental and Capiz, respectively. On the other hand, mussel production increased from 20.1MT in 2007 to 23.01MT in 2008; the Provinces of Capiz and Cavite have the highest production volume and showed the highest percentage increase (BAS, 2008).

In talaba and tahong production, there are many factors to be considered, such as physicochemical parameters, microbial load, and heavy metal content of the water. These water parameters may affect the quality of the oyster and green mussels produced in a particular farm and therefore affect the sustainability of the industry as a whole. A study conducted by Cebu and Orale (2017) on the physical and chemical properties of the Philippine mussel farms in Samar province, Philippines, found a variation in the physicochemical characteristics of the three Bays (Maqueday Bay, Kanbatutay Bay, and the Bay of Villareal). This variation was credited to the volume of water streaming into the cove and the bay’s physical variables. They indicated that water quality from 2004 to 2013, as monitored by 92 inhabitants, was at its most noticeably awful state, especially within the years 2007 and 2008, and has further worsened since then. Daniels et al. (2008) define water quality as the chemical, physical, and organic characteristics of water, and its appropriateness for assigned utilization. Each of these assigned employments has diverse characteristics; chemical, physical, and organic benchmarks essential to back the required need, for illustration, limits on the concentration of harmful substances or limitations on temperature and pH ranges for supporting oceanic communities.

According to Bhaja and Kundu (2012), water quality is affected by a wide range of natural and human influences. Natural influences include geological,

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 18

hydrological, and climatic. The impacts of human activities on water quality are both extensive and diverse, to the point that they destroy the natural flow and limit water use. The Philippine water quality standard on seawater for shellfish is called SA Water Fishery Class 1 (PHILMINAQ, n.d.). This classification of water quality is assigned to waters appropriate for the production, survival, and gathering of shellfish for household and industrial consumption. For example, the pH of seawater needed for marine life to flourish is near 8.3, and the dissolved oxygen must reach 5.0 mg/L. When contaminated, water affects aquatic ecosystems and fisheries decreases agricultural productivity, and causes diseases such as cholera, typhoid, hepatitis, diarrhea, typhoid, gastroenteritis, and dysentery. Therefore, water quality indicators can measure if water is appropriate for a certain purpose (The Aquarium of the Pacific, 2009).

Therefore, assessment and monitoring of Pilar Bay is of utmost significance to determine its water quality with regard to its physicochemical parameters. Further, geotagging of the existing oyster and mussel farms in Pilar Bay will be conducted. Hence, this research generally aimed to assess the environmental characteristics and determine the zonation of oyster and mussel culture areas within Pilar Bay, Capiz. Specifically, it aimed to (1) Geotag the farm areas of oyster and mussels among seven (7) sites within Pilar Bay, Capiz ; (2) Conduct monthly water physicochemical assessments and determine if there is a significant difference between sampling sites and between the months of sampling; and Determine the site with optimal water quality for oyster and mussel culture. In addition, this research mended this knowledge gap to explore the issues about the water quality in the oyster and mussel farms in Pilar Bay and thereby develop preventive measures if necessary.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Study site

The study site of the study was located in Pilar Bay, which encompasses the coastline of the Municipalities of Pilar, Pontevedra, Panay, President Roxas, and the City of Roxas. The selection of the site was based on the natural presence of bivalves such as oysters and mussels. Seven (7) sampling locations were determined based on the existing oyster and mussel culture farms in the area shown in Table. 1 and its corresponding plots in maps plotted from QGIS software (Fig. 1).

Table 1. Sampling sites, sampling stations, and its Central coordinates

Sampling sites	Sampling stations	Coordinates	
		Longitude	Latitude
Binaobawan, Pilar	1	122.952103° N	11.480238° E
	2	122.958687° N	11.47689° E
	3	122.861313° N	11.439742° E
San Pedro, Pontevedra	4	122.8711027° N	11.4360718° E
Madulano, Pres. Roxas	5	122.914935° N	11.457043° E
Pinamihagan, Pres. Roxas	6	122.914118° N	11.473268° E
	7	122.934766° N	11.47479° E
Agojo, Panay	8	122.828857° N	11.59702° E
	9	122.820904° N	11.603421° E
Buntod, Panay	10	122.828551° N	11.597249° E
	11	122.827512° N	11.594346° E
Bantigue, Panay	12	122.9010446° N	11.4921743° E
	13	122.9007377° N	11.492012° E

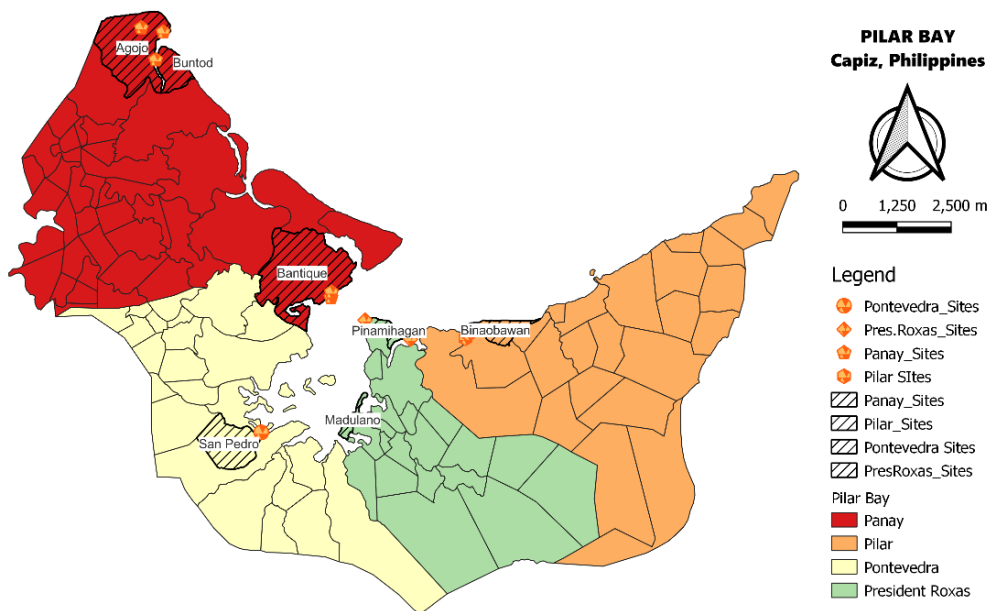


Figure 1. Location of the sampling stations in Pilar Bay, Capiz, Philippines

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 18

GPS Coordinates of the mussel and oyster farms from each site were collected using PDRP Geocamera. Coordinates were plotted on a map through open-source Geographic Information System software QGIS.

Evaluation of Physicochemical parameters of water

In situ, physicochemical parameters such as temperature (°C), Salinity (% Salt), Dissolved Oxygen (mg. L), pH, and Water Conductivity were measured at the water surface using RCYAGO 6 in 1 pH/RH/EC/CF/TDS(PPM)/Temperature Rechargeable Multi-Parameter Wall-Mount Water Quality Tester. Sensors were calibrated every sampling period. Monthly sampling was done every third week of the month from January until October 2023.

Rating scale according to Tan and Ransangan (2016) with its corresponding weighted value of physicochemical parameters applied for assessing the suitability of Pilar Bay, Capiz for oyster and mussel culture.

The optimum level set for mussels in terms of temperature ranged from 27-30 °C: salinity 27-35 ppt; pH 7.7-8.4 and Dissolved Oxygen >5 mg/L. Meanwhile, the optimum level set for oysters in terms of temperature ranges from 27-30 °C: salinity 17-26 ppt: pH 7.7-8.4 and DO >5 mg/L.

Commodities	Temperature (°C)	Salinity (ppt)	pH	Dissolved oxygen (>5 mg/L)
Mussel	27-30	27-35	7.7-8.4	>5 mg/L
Oyster	27-30	17-26	7.7-8.4	>5 mg/L
Weighted value	3	3	2	2

The suitability categories of sites for oyster and mussel farming were adapted from Sallih, 2005; Salmon, and Kingzett, 2002; Tan and Ransangan, (2016) with slight modifications on the remarks and the rating, points used.

Rating point	Site suitability	Remarks
1.0-3.0	Poor	The site is not suitable for oyster and mussel culture.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 18

3.1-7.0	Moderate	The site is moderately suitable for oyster and mussel culture.
7.1-10.0	Excellent	The site is highly suitable for oyster and mussel culture.

Statistical Analysis

Statistical Analysis was performed with IBM SPSS Statistics version 26 statistical software package. The data average of each parameter on each site was calculated using Microsoft Excel. Comparison between months and on each site for physicochemical parameters was performed using one-way analysis of variance (ANOVA) with a 95 % level of confidence. This is followed by a Tukey's Honestly Significant Difference post hoc comparison of pairwise differences, wherever a significant group difference was found.

(12) Results and Discussions (This part contains a presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Oyster and Mussel Farms in Pilar Bay

Pilar Bay which encompasses the coastline municipalities of Pilar, Pontevedra, Panay, President Roxas, and the city of Roxas has been one of the sources of income and livelihood of the fishing community in the area. Oyster and Mussel is one of the important commodities being cultured in the area. Thus, the importance of geotagging the existing mussel and oyster farms in Pilar Bay. (Figure 2) shows the map of geotagged mussel and oyster culture in Pilar Bay. In the municipality of President Roxas, barangay Pinamihagan shellfish growers use the raft method while barangay Madulano practices staking and broadcasting. In Panay, existing mussel farms in stake method are abundant in Barangay Buntod as well as the existing stake culture method of oyster farms in Agojo.

	Document Type:	DOCUMENTED INFORMATION	
		ISO 9001:2015	
	Document Title:	COMPLETED RESEARCH	
		Document Code	RES- CR-01
		Revision No.	00
		Effective Date	
		Page	Page 8 of 18

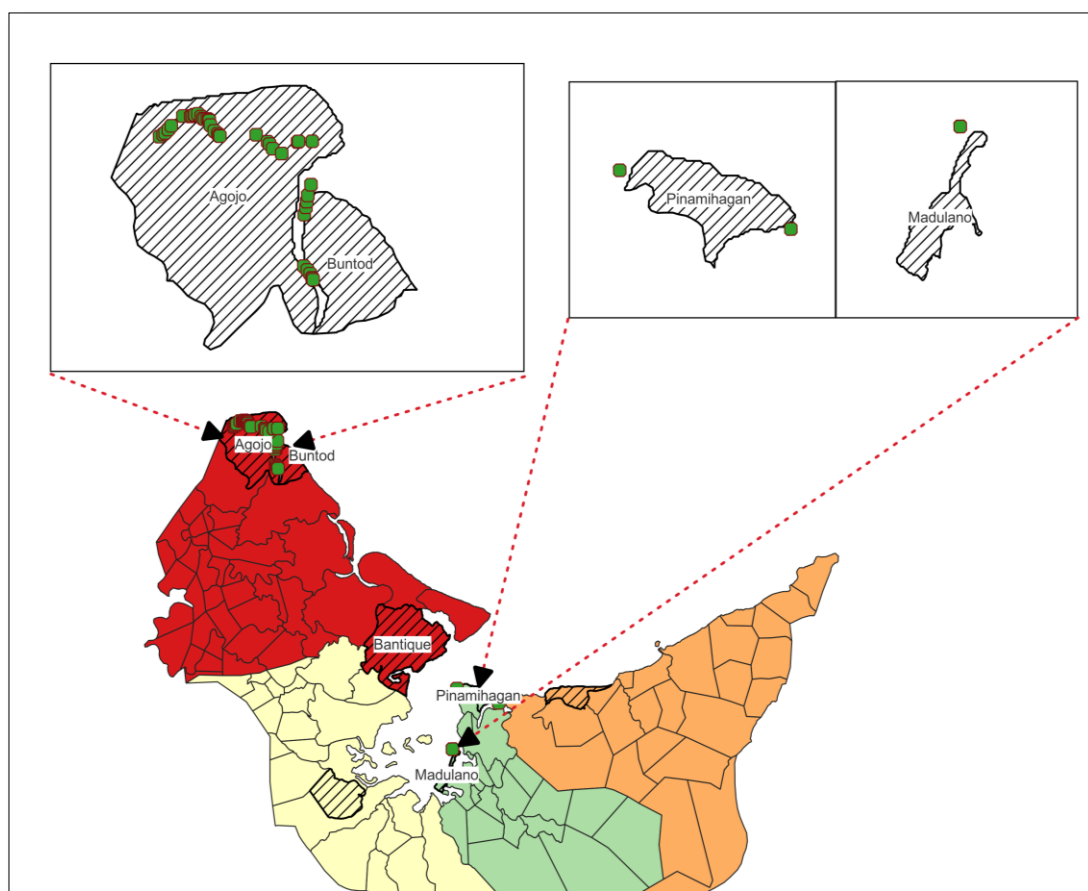


Figure 2. Map of existing Oyster and Mussel Farms in Pilar Bay

The temperature of Water in Pilar Bay based on Months

The physical water parameter of Pilar Bay in terms of temperature measured in °Celsius based on sites is shown in Figure 3.

The mean Temperature was significantly ($p=0.054$) lower in July in comparison to other sampling months.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:		Revision No.	00
		COMPLETED RESEARCH	Effective Date	
			Page	Page 9 of 18

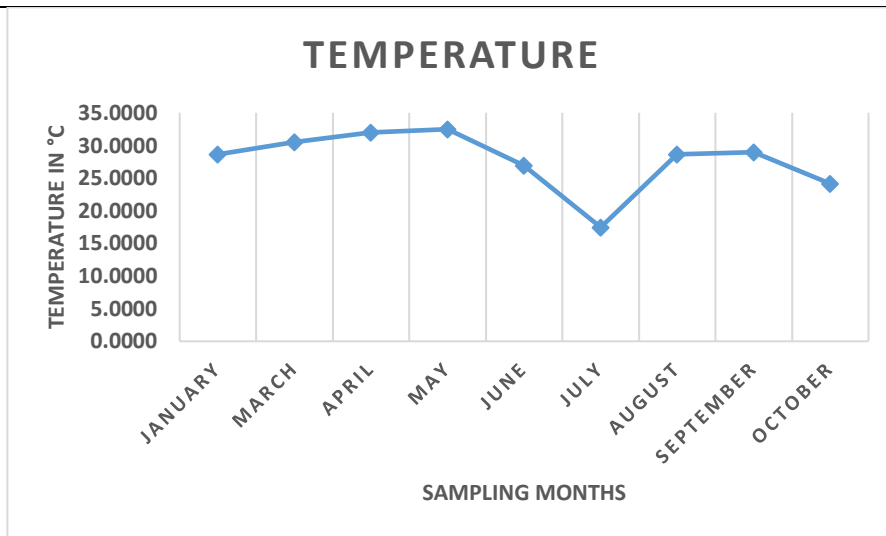


Figure 3. Physico-chemical water parameter of Pilar Bay in terms of Temperature measured in ° Celsius (°C)

The pH of Water in Pilar Bay based on Months

The chemical water parameter of Pilar Bay in terms of pH based on sites is shown in Figure 4.

Results revealed that the mean pH of July was significantly more basic compared to the rest of the month, except for June and October.

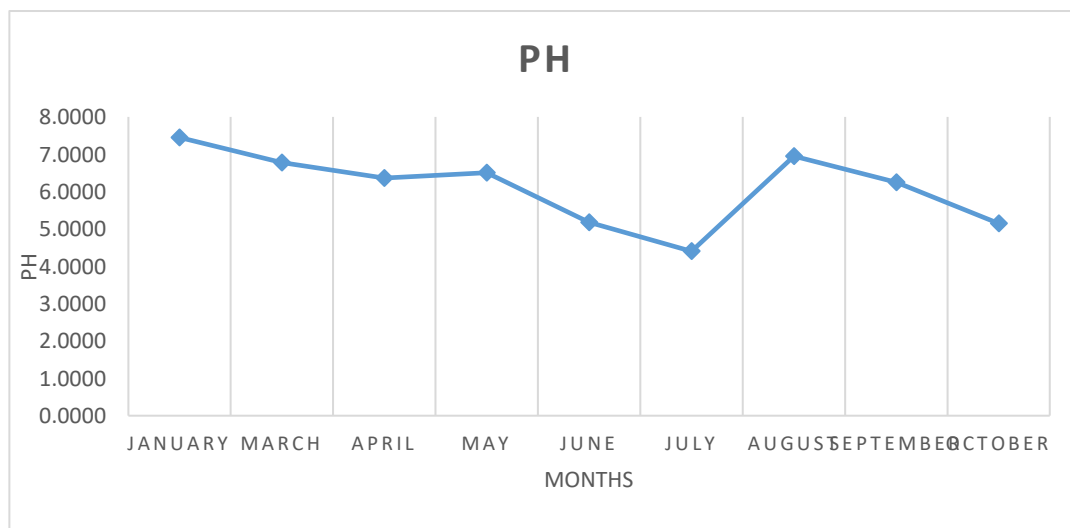


Figure 4. Physico-chemical water parameter of Pilar Bay in terms of pH

Dissolved Oxygen of Water in Pilar Bay based on Months

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 18

The chemical water parameter of Pilar in terms of dissolved oxygen (DO) in milligrams per liter (mg/L) based on month is shown in Figure 4.

The month of August had the highest dissolved oxygen (DO). Significant differences based on a 5% level of significance can only be seen in August.

Oysters and Mussel have the same requirement for the concentration of oxygen dissolved in water. The desirable range was > 5 mg/L (Layugan et al., 2018). The average result of the study ranges from 7-9 mg/L therefore it is in the desirable range for oyster and mussel culture.

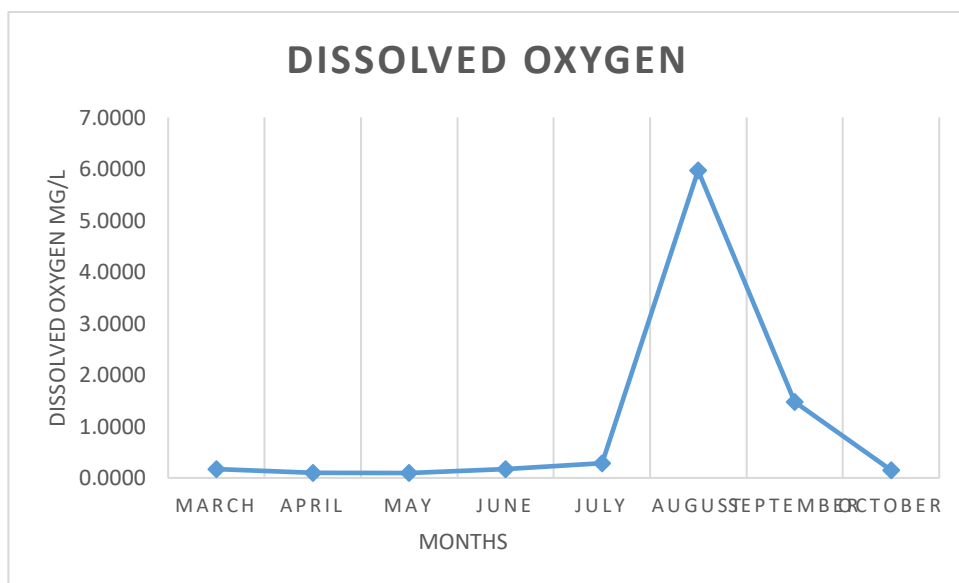


Figure 5. Physicochemical water parameter of Pilar Bay in terms of Dissolved Oxygen (mg/L)

Salinity of Water in Pilar Bay based on Months

The chemical water parameter of Pilar Bay in terms of salinity in parts per thousand (ppt) based on months is shown in Figure 6.

The result showed that May is significantly highly saline compared to other months. In contrast, mean salinity is significantly lower ($p=0.056$) in June and October.

The optimum salinity range for oysters is between 17-26 ppt meanwhile optimum salinity range for mussels is between 27-35 ppt (Layugan et al., 2018). Thus, the result in the pH of Pilar Bay waters varies within the ideal pH range for oyster and mussel culture.

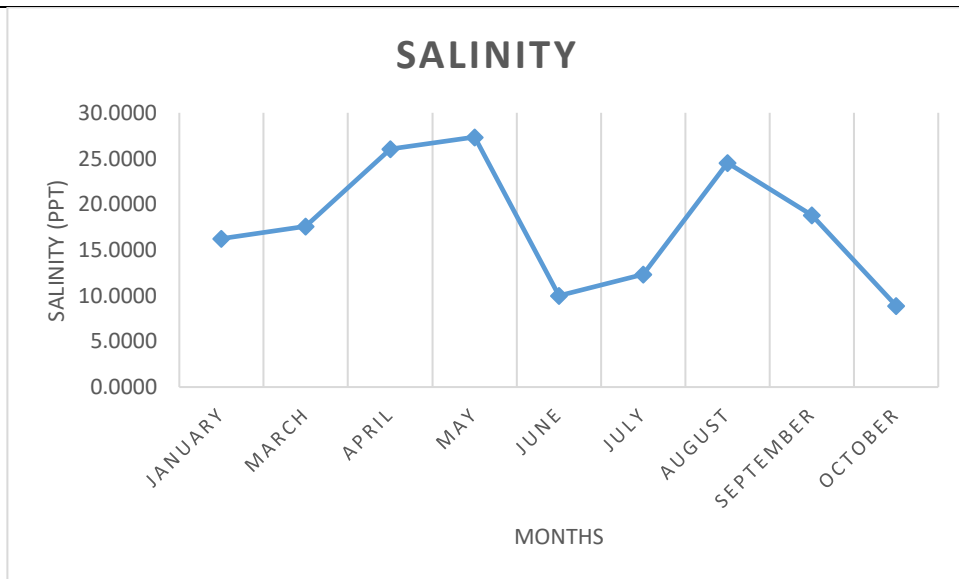


Figure 6. Physico-chemical water parameter of Pilar Bay in terms of Salinity (ppt)

Conductivity of Water in Pilar Bay based on Months

The chemical water parameter of Pilar Bay in terms of conductivity based on sites is shown in Figure 7.

The result revealed that April and May have significantly high conductivity. Meanwhile, the month of October has significantly recorded low water conductivity.

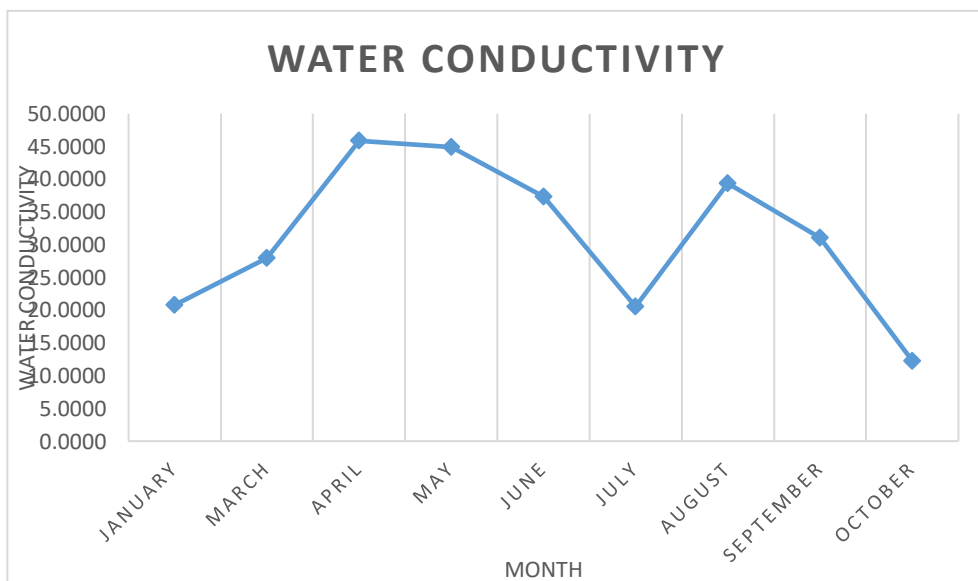


Figure 7. Physico-chemical water parameter of Pilar Bay in terms of Conductivity

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 12 of 18

Oxidation Reduction Potential (ORP) of Water in Pilar Bay Based on Months

The chemical water parameter of Pilar Bay in terms of ORP based on sites is shown in Figure 8.

The result shows that May has a significantly high oxidation-reduction potential and in contrast, month of July has a significantly low oxidation-reduction potential.

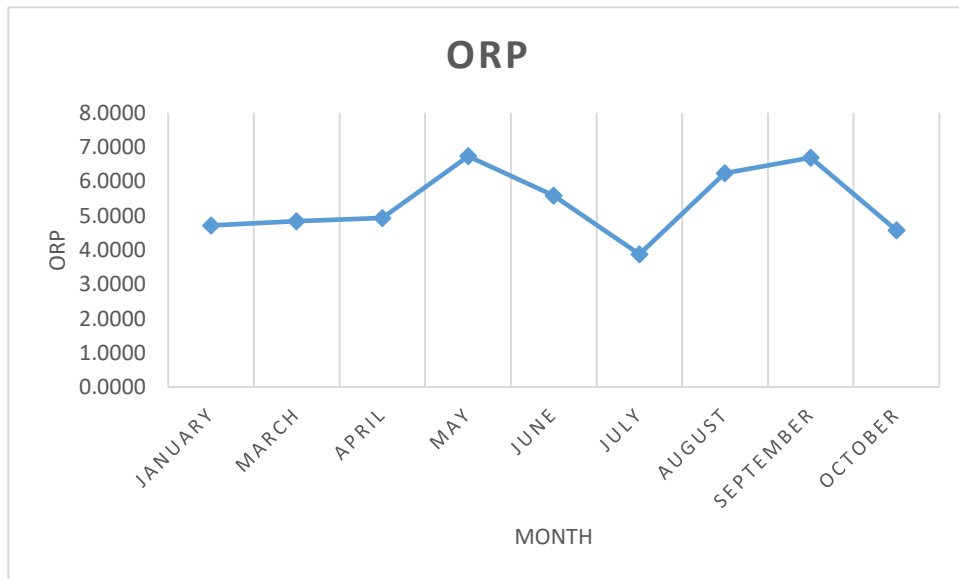


Figure 8. Physico-chemical water parameter of Pilar Bay in terms of ORP

Analysis of Variance on the physico-chemical water parameters of Pilar Bay in terms of Months shown in Table 1 revealed that selected physico-chemical parameters vary significantly with all p-values < 0.05 in terms of month.

Table 1. Analysis of Variance on the physicochemical water parameters of Pilar Bay in terms of Months

		ANOVA				
		Sum of Squares	df	Mean Square	F	Sig.
Temperature	Between Groups	2092.687	8	261.586	3.124	.003
	Within Groups	8625.909	103	83.747		
	Total	10718.597	111			
pH	Between Groups	97.108	8	12.139	2.483	.017

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 13 of 18

	Within Groups	503.476	103	4.888		
	Total	600.584	111			
DO	Between Groups	14673.888	8	1834.236	1231.481	.000
	Within Groups	153.414	103	1.489		
	Total	14827.301	111			
Salinity	Between Groups	4688.578	8	586.072	4.882	.000
	Within Groups	12365.316	103	120.052		
	Total	17053.894	111			
Conductivity	Between Groups	13686.927	8	1710.866	7.496	.000
	Within Groups	23508.903	103	228.242		
	Total	37195.830	111			
ORP	Between Groups	100.713	8	12.589	1.709	.105
	Within Groups	758.634	103	7.365		
	Total	859.346	111			

Physico-chemical water parameters of Pilar Bay based on Sites

Analysis of variance on the physicochemical water parameters of Pilar Bay based on Sites shown in Table 2. In terms of Temperature, there was a significant difference between each site ($p=0.001$). The mean temperature was significantly higher in Madulano (32.29°C), Binaobawan (31.29°C), and San Pedro (30.84°C) while the temperature was lowest at Bantigue (18.73°C).

In terms of pH, the result shows that there is a significant difference in all sites ($p=0.001$). The mean pH was significantly lower in Bantigue. Significant interaction was due to low variation in relation to other culture areas. Throughout the month of sampling, pH varies from 6.2 to 7.0.

In terms of Dissolved Oxygen, still there is no significant difference in all sites ($p=1.000$). In terms of salinity, results reveal that there is no significant difference with p-value ($p=0.124$). On the other hand, in terms of conductivity, results showed that water conductivity was significantly higher in Agojo and Buntod (43.11 and 43.93) while it was significantly lower in Bantigue (14.83). In terms of ORP, there is a significant difference between sites with p-value ($p=0.964$). Mean Oxidation-reduction potential was significantly higher in Pinamihagan (6.60) and significantly lower in Bantigue (3.31).

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 14 of 18

ANOVA						
		Sum of Squares	df	Mean Square	F	Sig.
Temperature	Between Groups	2057.057	6	342.843	4.156	.001
	Within Groups	8661.540	105	82.491		
	Total	10718.597	111			
pH	Between Groups	113.282	6	18.880	4.068	.001
	Within Groups	487.302	105	4.641		
	Total	600.584	111			
DO	Between Groups	36.834	6	6.139	.044	1.000
	Within Groups	14790.467	105	140.862		
	Total	14827.301	111			
Salinity	Between Groups	1580.958	6	263.493	1.717	.124
	Within Groups	16108.841	105	153.418		
	Total	17689.799	111			
Conductivity	Between Groups	11193.790	6	1865.632	7.534	.000
	Within Groups	26002.040	105	247.638		
	Total	37195.830	111			
ORP	Between Groups	110.689	6	18.448	2.587	.022
	Within Groups	748.657	105	7.130		
	Total	859.346	111			

Determination of site with optimal water quality for oyster and mussel culture

Site selection plays a critical role in determining the productivity, environmental impact, and interactions of aquaculture activities with ecosystem services. The selection of suitable sites affects both success and sustainability and it can be an important factor in production rates, mortality factors, environmental pollution, etc. (Amangabara et al., 2017).

Table 3 Shows the Site suitability of oysters based on their physicochemical parameters. The sampling sites of Madulano site 1, and Agojo site 1 are not suitable

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 15 of 18

for oyster culture. While, the sampling sites of Binaobawan site 1 and site 2, San Pedro site 1 and site 2, Pinamihagan site 1 and site 2, and Buntod site 1 and site 2 are moderately suitable for oyster culture. Moreover, the sampling sites of Bantigue site 1 and site 2 are highly suitable for oyster farming.

Table 3. Sites suitability based on physicochemical parameters for Oyster culture

Sampling Sites		Temperature (°C)	Salinity (ppt)	pH	DO	Total	Remarks
Binaobawan	Site 1	0	3	0	2	5	Moderate
	Site 2	0	3	0	2	5	Moderate
San Pedro	Site 1	3	0	0	2	5	Moderate
	Site 2	3	0	0	2	5	Moderate
Madulano	Site 1	0	0	0	2	2	Poor
Pinamihagan	Site 1	0	3	0	2	5	Moderate
	Site 2	0	3	0	2	5	Moderate
Bantigue	Site 1	3	3	0	2	8	Excellent
	Site 2	3	3	0	2	8	Excellent
Buntod	Site 1	3	0	0	2	5	Moderate
	Site 2	3	0	0	2	5	Moderate
Agojo	Site 1	0	0	0	2	2	Poor
	Site 2	3	0	0	2	5	Moderate

Table 4 Shows the Site suitability of mussels based on their physicochemical parameters. The sampling sites of Binaobawan site 1 and site 2, Madulano site 1, and Pinamihagan site 1 and site 2 are not suitable for mussel farming. While, the sampling sites of San Pedro site 1 and site 2, Bantigue site 1 and site 2 and Agojo site 1 are moderately suitable for mussel farming. However, the sampling sites of Buntod site 1 and site 2 and Agojo site 2 are highly suitable for mussel farming.

Table 4. Sites suitability based on physicochemical parameters for Mussel culture

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:		Revision No.	00
		COMPLETED RESEARCH	Effective Date	
			Page	Page 16 of 18

Sampling Sites		Temperature (°C)	Salinity (ppt)	pH	DO	Total	Remarks
Binaobawan	Site 1	0	0	0	2	2	Poor
	Site 2	0	0	0	2	2	Poor
San Pedro	Site 1	3	0	0	2	5	Moderate
	Site 2	3	0	0	2	5	Moderate
Madulano	Site 1	0	0	0	2	2	Poor
Pinamihagan	Site 1	0	0	0	2	2	Poor
	Site 2	0	0	0	2	2	Poor
Bantigue	Site 1	3	0	0	2	5	Moderate
	Site 2	3	0	0	2	5	Moderate
Buntod	Site 1	3	3	0	2	8	Excellent
	Site 2	3	3	0	2	8	Excellent
Agojo	Site 1	0	3	0	2	5	Moderate
	Site 2	3	3	0	2	8	Excellent

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

This study of environmental characteristics of water among oyster and mussel areas in Pilar Bay, Capiz showed that in terms of the physicochemical parameters of water significantly differed both in terms of months and based on sites except for Dissolved Oxygen and Salinity which was not significantly different in terms of sites. Temperature, according to the study of Ramos et.al.,2014 reproductive development of *C.gasar* and abundance and survival of larvae of *C. rhizophorae* and *C. gasar* (Lemos et al. 1994; Christo and Absher 2006) increase with temperature. Salinity variation in terms of months could be attributed to freshwater influx due to the frequent precipitation and flooding and tidal variation could influence variation in salinity. Dissolved oxygen concentrations were usually lower than the optimum limit of 5 mg/L. Low DO values may be due to increased runoff and higher turbidity (Asp et al. 2013) resulting in lower photosynthetic activity. Low dissolved oxygen concentrations among

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 17 of 18

culture units in our study may be cause for concern as low values, even for short periods, are negatively associated with growth, filtration ability, and disease susceptibility in oysters (Jeppesen et al. 2018; Breitburg et al. 2015). In addition, fluctuations in the pH of coastal and estuarine waters are large, being predicted to increase with ocean acidification, and increasingly lower seasonal values may disrupt oyster reproduction (Boulais et al. 2017). pH has been positively correlated with dissolved oxygen concentrations at oyster culture units due to the association between CO₂ production and O₂ consumption during respiration (Breitburg et al. 2015).

In addition, based on the result we can conclude that the following sampling sites are highly suitable for oyster farming; (i) Bantigue site 1 and site 2., the following sampling sites are moderately suitable for oyster farming (i) Binaobawan site 1 and site 2, (ii) San Pedro site 1 and site 2, (iii) Pinamihagan site 1 and site 2, (iv) Buntod site 1 and site 2. Moreover, the following sampling sites are not suitable for oyster farming (i) Madulano site 1 and (ii) Agojo site 1. On the other hand, the following sampling sites are highly suitable for mussel farming (i) Buntod site 1 and site 2 and (ii) Agojo site 2., the following sampling sites are moderately suitable for mussel farming (i) San Pedro site 1 and site 2, (ii) Bantigue site 1 and site 2, and (iii) Agojo site 1. Moreover, the following sampling sites are not suitable for mussel farming (i) Binaobawan site 1 and site 2, (ii) Madulano site 1, and (iii) Pinamihagan site 1 and site 2.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 18 of 18

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